Muhammad Shaheer Khan

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Summary

Final-year Computer Engineering student at METU with industry internship experience in machine learning and computer vision. Hands-on experience in feature engineering, imbalanced classification, YOLOv8 object detection, and model export using OpenVINO and TensorRT.

Education

Middle East Technical University (METU) | Ankara, Türkiye
B.Sc. Computer Engineering | Sep 2022 - Present | Expected graduation May-June 2027

Current coursework: Image Processing, Introduction to Robotics, Introduction to Computer Graphics, Software Construction, Senior Design Project.

Experience

Machine Learning Intern | Optiway Solutions | Ankara | Summer 2026

• Curated a transformer dissolved gas analysis dataset, removing duplicates and correcting mislabeled/misaligned records; retained 2,387 of 2,825 samples for model development.

• Engineered gas-ratio features for seven-class classification, improving XGBoost accuracy from 81.5% to 90.4% and Random Forest from 81.9% to 89.7%.

• Compared SMOTE, Borderline-SMOTE, ADASYN, CTGAN and TVAE using macro F1 and class-wise performance.

• Tuned Random Forest and XGBoost with GridSearchCV/RandomizedSearchCV and stratified five-fold cross-validation; evaluated held-out test performance.

Computer Vision Intern | Aremak | Ankara | Summer 2025

• Trained YOLOv8 models for real-time people detection on a custom dataset, achieving 0.91 mAP and inference speeds above 35 FPS.

• Compared variants using precision, recall, mAP and confusion matrices to assess accuracy/speed trade-offs.

• Exported trained models to OpenVINO and TensorRT for deployment on edge devices.

Projects

HOOPS - Basketball Video Analysis | Ongoing senior design project | 2026-2027

• Preparing an offline analysis system for recorded, fixed-camera basketball footage, with planned player/ball detection, tracking, court mapping and analytics.

• Prepared the project development plan, including staged delivery priorities, module integration, team coordination and evaluation planning.

Technical Skills

Programming: Python, C++, C, SQL
Machine Learning: scikit-learn, XGBoost, Random Forest, imbalanced-learn, SDV (CTGAN, TVAE)
Computer Vision: YOLOv8, object detection, dataset preparation, model evaluation
Data Analysis: NumPy, pandas
Deployment: OpenVINO, TensorRT
Tools: Git, Jupyter Notebook, Google Colab, VS Code; Linux (basic)

Languages

English - Fluent | Urdu - Native | Turkish - A2